

JOÃO VIANA

Data Scientist

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SUMMARY

I spent my PhD teaching computers to describe the universe's first fractions of a second. Now I point the same curiosity at data problems closer to home. Data holds a lot of value, though only as much as the processing, analysis and modelling behind it, and that is the part I enjoy most.

PROJECTS

Impact of AI on Students

EDA, regression on post-semester GPA, and threshold-tuned classification of burnout risk on a synthetic Kaggle dataset.

Skills: python (numpy, matplotlib, pandas, pytorch, scikit-learn)

Anomaly detection on the MVTec AD database

Zero-shot defect detection comparing convolutional autoencoders against PatchCore.

Skills: python (numpy, matplotlib, pytorch), patchcore

RAG-Powered NumPy Documentation Assistant

A sub-1B-parameter LLM boosted with a ChromaDB retrieval layer over the NumPy documentation, deployed with Docker Compose.

Skills: python (ollama, fastapi, streamlit, thread), docker/docker compose

SKILLS

- Maths: differential/integral calculus, linear algebra, statistics, multivariate analysis, complex analysis
- Coding: c++, python (matplotlib, numpy, pandas, scipy, thread, beautifulsoup, scikit-learn, pytorch), R, git, unittests, CI/CD, docker/docker-compose,
- Numerical Methods: ODE/PDE solver, minimisation algorithms, integration, Monte-Carlo integration
- OS: Linux, macOS, Windows
- Languages: Portuguese, English

RESEARCH

J. Viana, C. H. van Dorp, A. Nunes, et al. — Controlling the pandemic during the SARS-CoV-2 vaccination rollout, Nature Communications 12, 3674 (2021).

Topic: International collaboration with the DGS/Ministry of Health to predict the number of COVID-19 hospitalisations and the impact of the vaccination rollout.

EDUCATION

PhD, Physics and Astronomy

2022 - present

- P. Basler, L. Biermann, M. Mühleitner, J. Müller, R. Santos, J. Viana — BSMPT v3: A Tool for Phase Transitions and Primordial Gravitational Waves in Extended Higgs Sectors, arXiv:2404.19037.
- M. Mühleitner, J. Plotnikov, R. Santos, J. Viana — A Deep Dive into Baryon Asymmetry — the C2HDM, arXiv:2606.04229.

MSc, Nuclear and Particle Physics (Final grade: 19)

2019 - 2022

- Thesis : Dark Matter, Gravitational Waves and Higgs invisible decays (Thesis grade: 20)

BSc, Physics (Final grade: 18)

2016 - 2019